Reviews in Context or in Weights? Audited Pilots and a Controlled Comparison

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Abstract

We study whether a small agent should reuse a review history through context or weight updates, asking when adaptation improves the reviewed behavior while preserving task validity. Two Qwen3-8B pilots motivate a controlled follow-up: a Lean reporting evaluation collapsed to a compile-outcome shortcut, while a writing evaluation learned planted editorial rules but lost valid submissions. At the writing endpoint, the live weights-minus-context gap was -1.2 percentage points (95% interval $[-4.0, 1.5]$); training on a shared edited archive yielded a held-out gap of $+13.7$ $[11.4, 16.0]$. These results support archive-specific adaptation, but establish neither a lasting review-count crossover nor retention of general capabilities. We specify a shared-review comparison against a compact playbook, with coverage gates, validity safeguards, and clustered inference; this revised experiment remains unexecuted.

1. Introduction

Edits can convey preferences that were absent from the original instruction: prior work infers natural-language preferences from user revisions and retrieves them for later requests [2]. Our question is where to store that information: in retrieved reviews, in a maintained instruction playbook, or in a model adapter trained on corrected outputs. Our conditional thesis is that weight adaptation may eventually outperform a strong context baseline on identical review evidence without materially degrading public-task completion.

Our pilots reveal two obstacles to interpreting such a comparison: a high reporting score can conceal missing event coverage, and a high conditional style score can conceal invalid submissions. For writing, we therefore change the primary comparison from separately evolving workflows to representations of a common review archive, while retaining the live loop as a secondary deployment test; Lean already used shared reviews. This draft contributes audited pilots and a protocol for the unresolved comparison; neither superiority over a playbook nor continual learning has been demonstrated.

2. Related work

Long-context experiments compare retrieved demonstrations with fine-tuning and find that the relative ranking depends on data scale and task; LoRA does not uniformly overtake in-context learning [1]. We extend this comparison to reviewer corrections under a fixed evidence pool and validity constraints, without assuming a universal threshold. CIPHER retrieves preferences inferred from edits [2], while Agentic Context Engineering organizes experience into a playbook maintained through incremental updates [3]. These approaches motivate a learned summary baseline stronger in representational scope than the pilots' raw-example retrieval, although its performance in our tasks has not yet been measured.

LoRA freezes pretrained weights and trains low-rank update matrices [4]; self-distillation work uses a demonstration-conditioned model as its own teacher for on-policy learning [5]. We propose separate supervised and distillation arms, treating retention as an empirical question rather than a guarantee of small adapters. Model critics can also invent bugs [6], and evaluation methodology recommends paired comparisons and accounting for clustered questions [7]; these observations motivate reviewer calibration and grouping repeated samples by their underlying task.

3. Completed pilot setup

Both pilots used Qwen3-8B with a no-feedback arm, a retrieved-example context arm, and rank-16 LoRA retrained from base on cumulative corrections, with four live rounds of 120 sessions per arm. Ten concurrent workers per arm supplied throughput, not independent experimental replicates; the writing context budget was 8,000 tokens, and retrieval used normalized prompt-token overlap with recency tie-breaking. In the writing pilot, agents used file operations, a public format check, and submission tools to produce stories of 120 to 350 words containing two required words, while an Inkling editor applied eight hidden style rules. Guide A prohibited "suddenly," "very," semicolons, and em dashes, and constrained narrative tense, opening length, dialogue-tag verbs, and capitalization. The private scorer combined regex/token checks with a pinned spaCy

tagger; S is the per-story fraction of applicable rules passed, excluding the dialogue-tag rule when no tags occur, and U indicates valid submission, so reported mean $Y = US$ assigns invalid submissions zero. Training replay replaced the final story-producing action with the edited story and regenerated subsequent tool results; the pilot trained assistant messages, whereas the revised protocol restricts loss to the corrected turn. Writing pilot training used learning rate 4×10^{-4}, batch four through 60 rows and 16 thereafter, and at least three epochs and 30 steps.

Writing evaluation used 96 held-out prompts with four stochastic samples per policy and seed, including 48 prompts in genres absent from training; neither development nor held-out prompts entered the intended training split. The final live LoRA comparison used three training seeds, earlier checkpoints one; 10,000 paired prompt bootstrap draws produced pointwise intervals conditional on the collected training pools, without simultaneous coverage across budgets. The Lean pilot instead froze 120 trajectories from 40 lemmas in 20 families and sampled five reports per trajectory at each policy checkpoint; its reported score was precision times recall over structured facts, not F1 or explanation quality. Under that frozen rubric, a failed final compilation or absence of compilation produces only event E4, while E5 denotes a local `sorry` and E6 denotes an admitted dependency; this distinction matters when interpreting aggregate accuracy.

4. Audited pilot results

4.1. Lean: the evaluation did not identify a crossover

The held-out truth sets contained 113 E4-only traces, six E5-only traces, and one E5-plus-E6 trace, with no proved or partial outcomes and no E1, E2, or E3 cases. Always reporting E4 scores $113/120 = 0.9417$; a post hoc rule reporting E4 after failure or no compile and E5 otherwise scores 0.9958, matching all 600 sampled headers at each of the four LoRA checkpoints. On the lone dependency-event trace, every LoRA omitted E6 in all five samples, whereas the base and final context policy detected both events, demonstrating a regression hidden by the aggregate score. This shortcut is observationally sufficient to reproduce the headers, not proof of the model's internal computation; five reports on one trace also provide only one dependency example, so the intended crossover claim is withdrawn.

4.2. Writing: style learning with a validity cost

At 15 accepted edits, weights followed 90.4% of rules on valid stories versus 74.2% for context, but the corresponding validity-adjusted scores were 77.0% and 70.0%, showing why the two metrics must remain distinct. Table 1 reports the full observed curve: budgets 15 to 60 count accepted edits, later budgets count review opportunities, and the number of usable supervised targets differs from the nominal budget. At 480 opportunities, weights scored 84.9% versus 86.1% for context, a -1.2 point gap with interval $[-4.0, 1.5]$; the interval includes zero, and the 30-edit comparison was also inconclusive. At this endpoint, conditional style compliance was 95.8% versus 89.1%, but validity was 88.5% versus 96.6%; across the live weights evaluations, short stories and missing required words occurred among invalid outputs. The endpoint shows no reliable live-workflow advantage for weights, and pointwise early gains do not establish a lasting crossover.

Budget	SFT rows	Context	Weights	Gap
15	15	70.0	77.0	+7.0
30	30	77.0	76.1	-0.9
60	60	75.5	88.5	+12.9
120	116	78.5	88.2	+9.6
240	226	82.9	89.4	+6.4
360	338	85.6	92.3	+6.7
480	451	86.1	84.9	-1.2

Table 1. Writing pilot US scores (percent) and weights-minus-context gaps (percentage points); the unadapted base scores 57.1%. Endpoint weights average three seeds; other rows use one. These are separate evolving live pools, with early offline checkpoints.

A separate control used the same Inkling editor with guides A and B, which oppose three rules, and trained three seeds per guide on edits of the same 460 base drafts: held-out weights scored 91.8% versus 78.2% for context using A's archive, a $+13.7$ point difference $[11.4, 16.0]$. On the three opposed rules, the A-adapter's A-style advantage over the B-adapter was $+80.3$ $[78.3, 82.3]$ points, and the reversed B-style comparison was $+83.1$ $[80.8, 85.4]$, with invalid submissions still assigned zero. Edited-target training exceeded unedited-target training by $+33.5$ $[31.8, 35.2]$ points on guide A, supporting learning from these edits rather than an explanation based solely on additional training on the original drafts. These controls concern planted, mechanically scored preferences and raw-example context; they do not establish literary quality, superiority over a learned playbook, or a review-count threshold on a shared archive.

Selecting the best of four valid unique drafts by the fraction of original tokens surviving the edit improved hidden-rule compliance by $+3.6$ $[2.8, 4.4]$ points over uniform choice, compared with $+10.3$ $[9.8, 10.9]$ for an oracle selector. Each of 96 prompts began with 16 base-policy samples, filtered for validity and deduplicated before averaging four-candidate subsets and uniform ties; scores concern original drafts, token survival ignores pure insertions, and the pilot performed selection rather than reward-based training.

5. Revised experiment: specified, not yet run

5.1. Common evidence and adaptation arms

Following the v3 proposal, the primary experiment will freeze reviews of the no-feedback policy and evaluate budgets of 15, 30, 60, 120, 240, 360, and 480, reserving each arm's own four-round live loop for a secondary comparison. We plan six arms: no feedback; BM25 review retrieval capped at 8,000 tokens; a delta-updated playbook capped at 2,000 tokens; cumulative corrected-target SFT; playbook-conditioned self-distillation; and SFT plus the playbook. The playbook author will see only reviews and will never receive the writing scorer's secret rules; the distillation arm will use the playbook-conditioned policy as teacher and the unconditioned policy as student on training-pool prompts. The primary contrast will be SFT minus playbook; cumulative SFT restarts from base at each budget, making it a joint-training reference rather than a sequential continual learner or a guaranteed performance upper bound. Planned controls include step-matched SFT at every budget and sequential LoRA, self-target, and shuffled-target SFT at 480; these distinguish optimization exposure, ongoing updates, and informative corrections. The planned Qwen3-8B recipe uses non-thinking generation, temperature 0.7, top-p 0.8, top-k 20, unique session seeds, and prompts within 32,768 tokens; these are protocol choices rather than claims about currently available provider defaults. Training will use rank 16, batch 16, at least three epochs and 30 steps, three seeds, and a learning rate selected on development data from $\{1.5, 4.7, 7.0\} \times 10^{-4}$, with corrected-turn loss and downloaded adapters.

5.2. Task and reviewer gates

Before freezing Lean evaluation, development gates require base proof success between 20% and 70%, an exit-code baseline at most 0.85, tool errors at most 2%, and reviewer exactness at least 95%; these thresholds are proposed acceptance rules. The design calls for at least 60 sealed dependency groups, approximately 20% deliberately unprovable lemmas, and a separately reported compiler-checked challenge set with 20 positives and 20 negatives per rare event, preserving the natural test distribution. Lean's revised primary outcome will be exact report-header match over five samples per frozen trace, with report-turn-only adapters; whole-episode proof success will separately test spillover, and explanations will not be counted as verified by a header score. Writing gates require each secret rule to be passed by at most 70% of base development stories, editor recall at least 0.9 per rule, and at most 20% of changed words in clean sentences; targets failing public format checks will be excluded. A frozen reviewer from another model family and a 30-case human check are planned, while writing will retain both conditional style and US, plus a validity safeguard of no more than three percentage points below the base policy.

5.3. Inference and remaining work

The planned primary endpoint is the 480-review SFT-minus-playbook contrast; paired bootstrap analysis will cluster by writing prompt or Lean dependency group, retaining repeated samples within their cluster and reporting training-seed variation separately. A crossover will mean the earliest measured budget whose simultaneous lower confidence band remains above zero through 480, subject to the writing validity safeguard; if no budget qualifies, no crossover will be reported within the tested range. Development-only power simulations, a frozen implementation of the simultaneous band, reviewer gates, and new sealed test pools must precede execution; the completed pilots do not supply power or calibration evidence for this revised comparison. We plan retention measurements on public-task validity, an IFEval subset, and KL divergence from the base model, while editor-specificity controls and the optional token-survival reward experiment remain distinct from the primary comparison.

6. Discussion and conclusion

These pilots motivate auditing outcome coverage and validity denominators: Lean's aggregate score concealed missing event types, while writing style gains coexisted with public-constraint failures. Neither pilot contains a completed human editor smoke test or a general retention evaluation, and the writing study lacks blind prose-quality assessment; consequently, claims about human preference, general competence, and long-term personalization remain outside this evidence. The shared-archive writing result motivates a stronger contextual comparator; failed Lean calibration warrants withholding a crossover claim until coverage and reviewer gates pass. Corrections changed mechanically scored writing behavior, but the budget at which weights reliably beat a learned playbook without unacceptable validity loss remains unknown.

References

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